

Reference [100] Notes

Section 2 (Stability of numerical methods for linear problems)

- Linear system (1D)

$$\bullet \partial_t u(t, x) = -A \partial_x u(t, x) \quad (1)$$

$$\bullet u \in \mathbb{R}^{N+1} + A \in \mathbb{R}^{(N+1) \times (N+1)}$$

- A is diagonalizable w/ real eig-vecs $\Rightarrow (1)$ is hyperbolic

- Radiative transfer (1D)

$$\bullet \partial_t \psi + \mu \partial_x \psi + \sigma_a(x) \psi = \frac{\sigma_s(x)}{2} \phi - \sigma_s(x) \psi$$

$$- \psi = \psi(t, x, \mu) \text{ \& } \phi = \phi(t, x) = \int_{-1}^1 \psi d\mu$$

$$- \psi(t=0, x, \mu) = \psi_{IC}(x, \mu) = \psi_{IC}$$

$$- \psi(t, x_L, \mu) = \psi_L(t, \mu) \text{ \& } \psi(t, x_R, \mu) = \psi_R(t, \mu)$$

$$- t \in \mathbb{R}_+, x \in [x_L, x_R], \text{ \& } \mu \in [-1, 1]$$

- Let $P_\ell: [-1, 1] \rightarrow \mathbb{R}$ be normalized Legendre polys

$$- \psi(t, x, \mu) \approx \psi_N(t, x, \mu) := \sum_{\ell=0}^N u_\ell(t, x) P_\ell(\mu)$$

- Applying a Galerkin projection on (2) yields

$$\bullet \partial_t u = -A \partial_x u - \sigma_a u + S \sigma_s u \quad (3)$$

$$\bullet (A)_{kj} := \int_{-1}^1 \mu P_k P_j d\mu \text{ \& } (S)_{kj} := \delta_{kj} - \frac{c_k c_j}{2} \quad (c_i := \int_{-1}^1 P_i d\mu)$$

- Random Wave Equation

$$\bullet \partial_t u(t, x, \xi) + a(\xi) \partial_x u(t, x, \xi) = 0$$

$$- u(t=0, x, \xi) = u_{IC}(x, \xi)$$

$$- u(t, x_L, \xi) = u_L(t, \xi) \text{ \& } u(t, x_R, \xi) = u_R(t, \xi)$$

- General poly chaos basis Functions (GPC)

$$- u(t, x, \xi) \approx \sum_{\ell=0}^N u_\ell(t, x) P_\ell(\xi)$$

- Stochastic-Galerkin (SG) yields

$$- \partial_t u(t, x) = -A \partial_x u(t, x), (A)_{kj} = \mathbb{E}[a P_k P_j] \quad (5)$$

- Lax-Friedrichs Method

• Finite difference

$$\circ \partial_t u(t, x) \approx (u_{i+1}^{n+1} - \frac{1}{2}[u_{i+1}^n + u_{i-1}^n]) / \Delta t$$

$$\circ \partial_x u(t, x) \approx (u_{i+1}^n - u_{i-1}^n) / 2\Delta x$$

- Discretize the Full linear problem (1)

• Numerical Flux (Lax-Friedrichs)

$$\circ F^*(u, v) = \frac{1}{2} (A(u+v) - \lambda_{\max}(v-u))$$

$$\rightarrow \circ \text{Semi-discrete solution } u_{j,k} := \frac{1}{\Delta x} \int_{x_j}^{x_{j+1}} u_k(t, x) dx$$

Discretize
only in
space

$$\circ \dot{u}_j(t) = -\frac{1}{\Delta x} [F^*(u_j(t), u_{j+1}(t)) - F^*(u_{j-1}(t), u_j(t))]$$

$$\circ \dot{u}_j = \frac{\lambda_{\max}}{2\Delta x} [u_{j-1} - 2u_j + u_{j+1}] - \frac{1}{2\Delta x} A(u_{j+1} - u_{j-1}) \quad (\text{w/out } F^*) \quad (6)$$

• Simplify (6) w/matrices

* Define $L^{(1)}, L^{(2)} \in \mathbb{R}^{N_x \times N_x}$ as

$$L_{j,j+1}^{(1)} = L_{j,j-1}^{(1)} = \frac{V}{2}, \quad L_{j,j}^{(1)} = 1 - V \quad \text{and}$$

$$L_{j,j+1}^{(2)} = \pm \frac{\Delta t}{2\Delta x}, \quad \text{where } V := \lambda_{\max} \frac{\Delta t}{\Delta x}$$

* Then

$$\dot{u}(t) = \frac{1}{\Delta t} [(L^{(1)} - I)u(t) - L^{(2)}u(t)A^T] =: F(u(t)) \quad (7)$$

Discretize
in
time

→ * With Forward Euler in time

$$u^{n+1} = L^{(1)} u^n - L^{(2)} u^n A^T \quad (8)$$

- Determine Stability (L^2) of the Full problem

• Discret Fourier ansatz

$$\circ u_{j,k}(t) =: u_k(t, x_j) = \sqrt{\frac{\Delta x}{N+1}} \sum_{\alpha=1}^N \sum_{\ell=1}^{N+1} \hat{u}_{\alpha\ell}(t) e^{i\alpha\pi x_j} e^{2\pi i \frac{\ell k}{N+1}} \quad (9)$$

• Collect basis functions

$$\circ E_x = (\sqrt{\Delta x} e^{i\alpha\pi x_j})_{j,\alpha=1}^{N_x} \quad \& \quad E_\mu = \left(\frac{1}{\sqrt{N+1}} e^{2\pi i \frac{\ell m}{N+1}} \right)_{\ell,m=1}^{N+1}$$

$$\circ u^n = E_x \hat{u}^n E_\mu^H \quad \text{where } E_\mu^H = \text{adj}(E_\mu)$$

• Then

$$u^{n+1} = L^{(1)} E_x \hat{u}^n E_\mu^H - L^{(2)} E_x \hat{u}^n E_\mu^H A^T$$

• Define $D^{(1)}$ & $D^{(2)}$ as such

$$\circ D_{\alpha\beta}^{(1)} = (1 - V + V \cos(\alpha\pi\Delta x)) \delta_{\alpha\beta}$$

$$\circ D_{\alpha\beta}^{(2)} = i \frac{4\epsilon}{\Delta x} \sin(\alpha\pi\Delta x) \delta_{\alpha\beta}$$

$$\circ \text{Then } L^{(1)} E_x = E_x D^{(1)} \text{ \& } L^{(2)} E_x = E_x D^{(2)} \quad (10)$$

• Thus,

$$\begin{aligned} u^{n+1} &= (E_x D^{(1)} \hat{u}^n E_\mu^H - E_x D^{(2)} \hat{u}^n E_\mu^H A^T) \\ &= E_x (D^{(1)} \hat{u}^n - D^{(2)} \hat{u}^n E_\mu^H A^T E_\mu) E_\mu^H \end{aligned}$$

• Since the Flux matrix is symmetric there is an orthogonal eig-decomp s.t. $T^T \Lambda T = E_\mu^H A^T E_\mu$ where $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_{N+1})$ of A^T .

$$\circ \text{Let } v^n = \hat{u}^n T^T \rightarrow \|v^n\|_F = \|\hat{u}^n\|_F = \|u^n\|_F$$

$$\circ u^{n+1} = E_x (D^{(1)} v^n - D^{(2)} v^n \Lambda) T E_\mu^H$$

• Then we have a bound!

$$\begin{aligned} \circ \|u^{n+1}\|_F &\leq \max_{\alpha,\beta} |D_{\alpha\alpha}^{(1)} - \lambda_\beta D^{(2)}| \|u^n\|_F \\ &= \sqrt{(1-V)^2 - 2(1-V)\cos(\alpha\pi\Delta x) + 1} \|u^n\|_F \\ &= \sqrt{V^2 - 2V(1 - \cos(\alpha\pi\Delta x)) + 2(1 - \cos(\alpha\pi\Delta x))} \|u^n\|_F \end{aligned} \quad (11)$$

• To get L^2 stability $V \leq 1$

A similar approach $\rightarrow$

• Doesn't prove convergence for non-linear problems

- Stability of scattering terms

• Back to radiative transfer

$$\circ G := \sigma_s S - \sigma_a I \quad (G \text{ is diagonal})$$

• Scattering ($G u(t, x)$) & Streaming ($-A \partial_x u(t, x)$)

are treated separately w/a semi-lag approach

$$\circ \partial_t u_I(t, x) = -A \partial_x u_I(t, x), \quad u_I(t_0, x) = u_0(x) \quad (12)$$

$$\circ \partial_t u_{II}(t, x) = G u_{II}(t, x), \quad u_{II}(t_0, x) = u_I(t_1, x)$$

- To solve each substep, we use the full discretized problem (i.e. eq (8)) on the linear advection-type PDE and a standard solver (like Forward Euler) For the second PDE

$$\begin{aligned} \circ u^{n+1/2} &= L^{(1)} u^n - L^{(2)} u^n A^T \\ \circ u^{n+1} &= u^{n+1/2} + \Delta t (u^{n+1/2})^T G \end{aligned} \quad (13)$$

- As a single step

$$u^{n+1} = (L^{(1)} u^n - L^{(2)} u^n A^T)^T (I + \Delta t G) \quad (14)$$

- For Error bounds use (11) directly

$$\circ \|u^{n+1}\|_F \leq \max_{\alpha} |\tilde{D}_{\alpha\alpha}| \|u^n\|_F \quad (15)$$

$$\text{where } \tilde{D}_{\alpha\alpha} = |D_{\alpha\alpha}^{(1)} - \lambda_{\max} D_{\alpha\alpha}^{(2)}| \max_{\ell} |1 - \Delta t G_{\ell\ell}|$$

- With $\nu = \lambda_{\max} \frac{\Delta t}{\Delta x}$ then

$$\|u^{n+1}\|_F \leq \max_{\ell} |1 + \Delta t G_{\ell\ell}| \sqrt{(1 - \nu \cos(\alpha \pi \Delta x))^2 + \nu^2 \sin^2(\alpha \pi \Delta x)} \|u^n\|_F$$

Section 3

- Use DLRA to represent and evolve the solutions of type (1) on a manifold of rank-r Functions

- Discretize First (i.e. eq (7))

- Let $u(t) \in \mathbb{R}^{N \times (N+1)}$ be the solution represented as

$$\circ u(t) \approx X(t) S(t) W(t)^T \in M_r \quad \leftarrow \begin{matrix} \text{set of} \\ \text{low rank} \\ \text{sol.} \end{matrix} \quad (16)$$

$$\circ X \in \mathbb{R}^{N \times r}, S \in \mathbb{R}^{r \times r}, \& W \in \mathbb{R}^{(N+1) \times r}$$

- Goal: Find equations For $X, S, \& W$

- In other words, we want $u_r \in M_r$ s.t.

$$u_r(t) \in T_{u_r} M_r \quad \text{s.t. } \|\dot{u}_r(t) - F(u_r(t))\| \text{ is minimized} \quad (17)$$

- F denotes the semi-discrete scheme From (7) and

$$T_{u_r} M_r \text{ is the tangent space of } M_r \text{ at } u_r(t).$$

$$\circ \text{By Lemma 4.1, } \dot{u}_r(t) = P(u_r(t)) F(u_r(t)) \quad (18)$$

◦ Where P is the orthogonal proj onto the tangent space, given as

$$P_g = XX^T g - XX^T g WW^T + g WW^T$$

◦ Use a Lie-Trotter splitting scheme on (8)

$$* \dot{u}_I(t) = F(u_I(t)) WW^T, u_I(t_0) = u_r(t_0) \quad \top$$

$$* \dot{u}_{II}(t) = -XX^T F(u_{II}(t)) WW^T, u_{II}(t_0) = u_I(t_1) \quad (19)$$

$$* \dot{u}_{III}(t) = XX^T F(u_{III}(t)), u_{III}(t_0) = u_{II}(t_1) \quad \perp$$

◦ Then $u_r(t_1) = u_{III}(t_1)$

◦ This completes the proj-splitter integrator

after reforming (19) with the subs

$$u_\alpha(t) \approx X_\alpha(t) S(t) W(t)^T = K(t) W(t)^T \text{ and } = X(t) L(t)$$

◦ The unconventional integrator (i.e. Bulg) was also mentioned

- Discretize Last

• Ansatz: $u_r(t, x) = \sum_{j=1}^n X_j(t, x) S_j(t) W_j(t)$

◦ $X_j: \mathbb{R}_+ \times \mathbb{R} \rightarrow \mathbb{R}$ & $W_j: \mathbb{R}_+ \rightarrow \mathbb{R}^{N+1}$ \top

◦ W_j are orthogonal, $W_j(t)^T W_j(t) = I$ (26)

◦ X_j are orthogonal, $\langle X_j(t, x), X_i(t, x) \rangle_x = \delta_{ij}$ \perp

• Split equation (where $u(t, x) = W(t) S(t) X(t, x)$???)

◦ $\partial_t u_I(t, x) = -WW^T A \partial_x u_I(t, x), u_I(t_0, x) = u_r(t_0, x)$ \top

◦ $\partial_t u_{II}(t, x) = P_x(W^T A \partial_x u_{II}(t, x)), u_{II}(t_0, x) = u_I(t_1, x)$ (27)

◦ $\partial_t u_{III}(t, x) = -P_x(A \partial_x u_{III}(t, x)), u_{III}(t_0, x) = u_{II}(t_1, x)$ \perp

◦ where $P_x g := \sum_{i=1}^n \langle g, X_i(t, x) \rangle_x X_i, X(t, x) = (X_j(t, x))_{j=1}^n$

and $W(t) = (W_j(t))_{j=1}^n$, i.e. $X \in \mathbb{R}^n$ & $W \in \mathbb{R}^{(n+1) \times n}$

• Discretize (with any grid)

Section 4

- Discrete DLRA

- Apply the numerical Flux (7) & Forward Euler (in time) to the K , S & L ODEs From (20), (21), & (22)

$$\circ K^{n+1} = K^n + [(L^{(1)} - I)u_I^n - L^{(2)}u_I^n A^T]W^n$$

$$\star K^{n+1} = X^{n+1} \tilde{S}^n \text{ (achieved w/QR decomp)}$$

$$\circ \tilde{S}^{n+1} = \tilde{S}^n - (X^{n+1})^T [(L^{(1)} - I)u_I^n - L^{(2)}u_I^n A^T]W^n \quad (29)$$

$$\circ L^{n+1} = L^n + (X^{n+1})^T [(L^{(1)} - I)u_{II}^n - L^{(2)}u_{II}^n A^T]$$

$$\star L^{n+1} = S^{n+1} (W^{n+1})^T$$

- Make $u_\alpha^n = X S W^T$ sub and simplify

$$\circ u_I^{n+1} = (L^{(1)}u_I^n - L^{(2)}u_I^n A^T)W^n$$

$$\circ u_{II}^{n+1} = -X^{n+1} (X^n)^T [(L^{(1)} - 2I)u_{II}^n - L^{(2)}u_{II}^n A^T]W^n \quad (30)$$

$$\circ u_{III}^{n+1} = X^{n+1} (X^n)^T (L^{(1)}u_{III}^n - L^{(2)}u_{III}^n A^T)$$

- Like in Sec 2, we use the Fourier Transform to get $u^n = E_x \tilde{u}^n (W^n)^T$, where $\tilde{u}^n \in \mathbb{R}^{N_x \times n}$ is the FT of u^n .

- Also, recalling everything in (30) is of the form $\tilde{u}^n = X S W^T$, we get the FT of u^n is

$$\circ \tilde{u}_I^n = E_x^H X^n S^n$$

$$\circ \tilde{u}_{II}^n = E_x^H X^{n+1} \tilde{S}^n$$

$$\circ \tilde{u}_{III}^n = E_x^H X^{n+1} \tilde{S}^{n+1}$$

- Finally using the FTs of $L^{(1)}$ & $L^{(2)}$ From (10) and defining $\tilde{A} := (W^n)^T A^T W^n$

$$\circ u_I^{n+1} = E_x (D^{(1)} \tilde{u}_I^n - D^{(2)} \tilde{u}_I^n \tilde{A}) (W^n)^T$$

$$\circ u_{II}^{n+1} = -X^{n+1} (X^n)^T E_x [(D^{(1)} - 2I) \tilde{u}_{II}^n - D^{(2)} \tilde{u}_{II}^n \tilde{A}] (W^n)^T \quad (31)$$

$$\circ u_{III}^{n+1} = X^{n+1} (X^n)^T E_x (D^{(1)} \tilde{u}_{III}^n - D^{(2)} \tilde{u}_{III}^n \tilde{A}) (W^n)^T$$

- Thm 1 - the proj-splitter w/(1) & (36) is L^2 -unstable.
 - Thm 1 shows that instability comes only from the S step in combination with the Lax-Friedrichs disc. scheme.

- Continuous DLRA

- Since K is the only variable with a PDE, numerical Fluxes are only needed for the K step and steps S and L can be solved as ODEs

◦ Assume the stencil $2_x \bar{X}_K(t_n, x) \big|_{x_j} = \frac{1}{2\Delta x} (\bar{X}_{j+1/2}^n - \bar{X}_{j-1/2}^n)$

◦ $K^{n+1} = L^{(1)} K^n - L^{(2)} K^n \tilde{A}$

◦ $\tilde{S}^{n+1} = \tilde{S}^n + (\bar{X}^{n+1})^T L^{(2)} \bar{X}^{n+1} \tilde{S}^n \tilde{A}$ (32)

◦ $L^{n+1} = L^n - (\bar{X}^{n+1})^T L^{(2)} \bar{X}^{n+1} \tilde{S}^{n+1} (W^{n+1})^T A^T$

• Thm 2 - w/CFL condition $\lambda_{\max}(\hat{A}) \frac{\Delta t}{\Delta x} \leq \frac{1}{\sqrt{3}}$,

the proj-splitting scheme (32) is

L^2 -stable, i.e., $\|U^{n+1}\|_F \leq \|U^n\|_F$.

- While the issue w/the S step has been solved, the CFL condition isn't very good in the schemes current set-up

— Consider the scheme below

- K^{n+1} (From Cont. DLRA)

- $\tilde{S}^{n+1}$ (From Disc. DLRA) where $[L^{(1)} - I] \rightarrow -[L^{(1)} - I]$

- L^{n+1} (From Disc. DLRA)

- In S-step, $[L^{(1)} - I]$ is the stabilization term and does effect "consistency", thus the sign change does not effect consistency

• Thm 3 - Thm 2 w/CFL cond. $\lambda_{\max}(\tilde{A}) \frac{16}{\lambda_8} \leq 1$